Course 1: Foundations of Project Management — Study Guide

Module 1: Embarking on a Career in Project Management

- **Project Management:** The application of knowledge, skills, tools, and techniques to meet project requirements and achieve the desired outcome. It enables organizations to deliver results efficiently by keeping work structured and aligned to goals.
- **Project:** A unique, temporary endeavor with a defined beginning and end, undertaken to create a specific product, service, or result. Unlike ongoing operations, projects conclude once their objectives are met.
- **Project Manager:** The person responsible for the day-to-day management of a project. They coordinate people, resources, and timelines to ensure the project is delivered successfully.

Core Skills of a Project Manager:

- **Coordination:** Getting people and teams to work together toward the end goal. This includes managing dependencies between tasks and ensuring nothing falls through the cracks.
- **Organization:** Staying focused on multiple concurrent tasks while maintaining attention to detail. Strong organizational skills prevent scope creep and missed deadlines.
- **Leadership:** Guiding a group of people through effective planning, task coordination, and key decision-making. Leadership also involves motivating team members and managing conflict constructively.

Additional Competencies:

- **Decision-Making:** Enabling the team and stakeholders to make timely, informed decisions — and knowing when to escalate issues to higher authority.
- **Communication:** Keeping all parties informed through regular updates, clear documentation, and active listening.
- **Flexibility:** Adapting plans when circumstances change, without losing sight of project goals.

Core Responsibilities: Planning and organizing, managing tasks, managing the schedule, controlling the budget, tracking issues and risks, managing quality, and overseeing project scope.

Cross-Functional Teams: Groups of people with different functional expertise (e.g., engineering, marketing, finance) working together toward a common project goal. The project manager bridges these disciplines to maintain alignment.

Module 2 & 3: The Project Management Life Cycle and Methodologies

The Project Management Life Cycle

A 4-phase framework that provides structure from a project's conception through its completion:

1. **Initiate the Project:** Define project goals, scope, and deliverables. Identify key stakeholders and team members, secure resource commitments, and obtain formal stakeholder approval before work begins.
2. **Make a Plan:** Establish the budget and schedule, assign roles and responsibilities, identify potential risks, and create a communication plan. A thorough plan serves as the project's single source of truth throughout execution.
3. **Execute the Project:** Monitor the team as they complete tasks, remove blockers, manage stakeholder expectations, and adapt the plan in response to changes or risks that materialize.
4. **Close the Project:** Confirm all deliverables have been accepted, formally release team members, hand off final outputs to the client or sponsor, document lessons learned, and celebrate team achievements.

Project Management Methodologies

Methodology	Core Principle	Best Used When
Waterfall	Linear, sequential phases; each must be complete before the next begins. PM leads and assigns tasks.	Requirements are stable and well-defined at the start; low likelihood of scope change.
Agile	Iterative, short work cycles (sprints) with frequent review and adaptation. Teams self-manage.	Requirements are expected to evolve; fast delivery of value and user feedback is critical.
Scrum	Agile framework using Sprints, a Scrum Master facilitator, and cross-functional teams.	Complex product development needing high collaboration and rapid iteration.
Kanban	Visual board (To Do / In Progress / Done) limits work-in-progress and exposes bottlenecks.	Continuous-flow work or support teams where work arrives unpredictably.
Lean	Eliminate the 8 wastes (defects, overproduction, waiting, inventory, transportation, motion, excess processing, non-utilized talent) using the **5S tool**: Sort, Set in Order, Shine, Standardize, Sustain.	Process-heavy environments seeking efficiency and cost reduction.
Six Sigma	Reduce variation to achieve 99.9996% defect-free output using **DMAIC**: Define, Measure, Analyze, Improve, Control.	Quality-critical industries where defect costs are high.
Lean Six Sigma	Combines Lean waste elimination with Six Sigma variation reduction.	Projects aiming to simultaneously cut costs, improve quality, and accelerate processes.

Waterfall vs. Agile — Detailed Comparison

Dimension	Waterfall	Agile
PM Role	Active director; prioritizes and assigns tasks.	Scrum Master / facilitator; removes blockers, not a task-assigner.

Scope	Fixed early; changes require formal change requests.	Evolves each sprint based on continuous stakeholder feedback.
Schedule	Sequential phases with set milestones.	Time-boxed Sprints (typically 2–4 weeks) with defined deliverables.
Cost	Controlled via upfront estimates and fixed budgets.	May shift per iteration; value delivered early offsets uncertainty.
Communication	PM drives stakeholder updates on a cadence.	Team communicates directly with users/ customers throughout.
Documentation	Extensive; serves as the blueprint for execution.	Lightweight; working software/deliverable is preferred over docs.

Module 4: Organizational Structure and Culture

Organizational Structure

The way a company is arranged — defining reporting lines, authority, and resource flow — directly impacts a project manager's ability to secure resources and make decisions.

- **Classic (Functional/Top-Down):** Authority flows from a CEO downward through VPs and department heads. PMs have limited authority and must negotiate resources through functional managers.
- **Matrix:** Employees report to both a functional manager and a project manager (or program stakeholder). PMs have more direct access to resources but must manage competing priorities from multiple leaders.
- **Authority:** A PM's authority — their ability to make binding decisions — is typically greater in a Matrix structure than in a Classic one.
- **Resource Availability:** Understanding how to access the people, equipment, and budget required. This knowledge is essential before committing to a project schedule.

Organizational Culture

The shared values, behaviors, rituals, and history that define how an organization operates — effectively the "company's personality." Culture can either accelerate or obstruct a project's progress.

How to Navigate Culture as a PM:

- **Ask questions** during interviews and onboarding — about communication norms, decision-making authority, risk tolerance, and how success is recognized.
- **Make observations** — notice how meetings are run, how disagreements are resolved, and what informal rituals exist.
- **Understand your impact** — recognize that your actions and communication style will either align with or challenge the culture, and adjust accordingly.

Key Cultural Dimensions to Assess:

- *Atmosphere:* Dress code, risk appetite, feedback norms, celebration of success.
- *Policies:* Flexibility (remote work, hours), leave policies, DEI practices.
- *Processes:* Onboarding procedures, how project impact is measured.
- *Values:* Mission statement alignment, investment in professional development.

Change Management

Project delivery often requires changes to existing processes, systems, or behaviors. Managing this human side of change is as important as the technical execution.

- **Change Management:** The process of delivering a completed project and getting people to adopt and use the change. Resistance is natural; the PM's role is to reduce friction.
- **Change Agent:** A person inside the organization who champions the transformation, focuses on improving organizational effectiveness, and helps others embrace the new way of working.
- **Core Principles:**
- Create a sense of **ownership and urgency** so stakeholders are invested in the outcome.
- Assemble the **right mix of skills and personalities** on the change team.

- Maintain **transparent, accessible communication** — be upfront with plans and make information available to all affected parties.
- **Adoption Tools:** Feedback mechanisms (surveys), flowcharts (visualize the future-state process), and culture mapping (illustrate how values and behaviors will shift).

Project Governance

The framework that defines how project decisions are made, who has authority, and how accountability is maintained.

- Governance covers policies, regulations, functions, processes, procedures, and responsibilities.
- It ensures projects remain on time, within budget, and aligned to organizational objectives.
- In a **Classic structure**, governance authority sits primarily with functional leaders. In a **Matrix structure**, governance is more distributed, requiring the PM to navigate multiple stakeholders.